



Ordinary Differential Equations

First-Order Differential Equations

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ENGR 213**

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Today's Lecture

- Exact Equations

Reference

Advanced Engineering Mathematics, D.G. Zill (7th edition), **Sections 2.4**

Exact Equations

- ❑ A differential expression $M(x, y) dx + N(x, y) dy$ is an **exact differential** in a region R of the xy -plane if it corresponds to the differential of some function $f(x, y)$.
- ❑ A first-order differential equation of the form

$$M(x, y)dx + N(x, y)dy = 0$$

is said to be an **exact equation** if the expression on the left side is an exact differential.

e.g.,: $x^2y^3dx + x^3y^2dy = 0$

□ **Theorem** - Criterion for an exact differential

Let $M(x, y)$ and $N(x, y)$ be continuous and have continuous first *partial derivatives* in a rectangular region R defined by $a < x < b$, $c < y < d$. Then:

A necessary and sufficient condition that $M(x, y) dx + N(x, y) dy$ be an exact differential is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

e.g.,: $x^2y^3dx + x^3y^2dy = 0$

□ Method of solution

- I. Given a differential equation of the form $M(x,y) dx + N(x,y) dy$, determine whether $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. If does, then there exist a function $f(x,y)$ such that

$$\frac{\partial f}{\partial x} = M(x, y), \quad \frac{\partial f}{\partial y} = N(x, y)$$

- II. From $\frac{\partial f}{\partial x} = M(x, y)$ find $f(x, y) = \left[\int M(x, y) dx \right] + g(y)$

- III. From $\frac{\partial f}{\partial y} = N(x, y)$ determine $g'(y)$:

$$\frac{\partial f}{\partial y} = \left[\frac{\partial}{\partial y} \int M(x, y) dx \right] + g'(y) = N(x, y) \rightarrow g'(y) = N(x, y) - \frac{\partial}{\partial y} \int M(x, y) dx$$

IV. By integrating with respect to y , find $g(y)$:

$$g(y) = \int N(x, y) dy - \int \frac{\partial}{\partial y} \int M(x, y) dx dy$$

V. Finally, substitute $g(y)$ in $f(x, y) = [\int M(x, y) dx] + g(y)$



EXAMPLE 1

Solve $2xy \, dx + (x^2 - 1) \, dy = 0$

EXAMPLE 2

Solve $\frac{dy}{dx} = \frac{xy^2 - \cos(x) \sin(x)}{y(1 - x^2)}, \quad y(0) = 2$

EXAMPLE 3 (a non-exact DE made exact)

Solve $xy \, dx + (2x^2 + 3y^2 - 20) \, dy = 0$

- If $(M_y - N_x)/N$ is a function of x alone, multiplying the DE by $e^{\int \frac{M_y - N_x}{N} dx}$ results in an exact equation.
- If $(N_x - M_y)/M$ is a function of y alone, multiplying the DE by $e^{\int \frac{N_x - M_y}{M} dy}$ results in an exact equation.

$$xy \, dx + (2x^2 + 3y^2 - 20) \, dy = 0$$